# MUSCLE RECOVERY & OVERTRAINING PREDICTION

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# INTRODUCTION

In sports and fitness, proper recovery is essential for improving performance. When athletes train excessively without enough rest, it leads to **overtraining syndrome**, which can reduce performance and increase injury risk.

Machine Learning helps analyze training and health data to **predict whether an athlete is recovered or at risk of overtraining**.

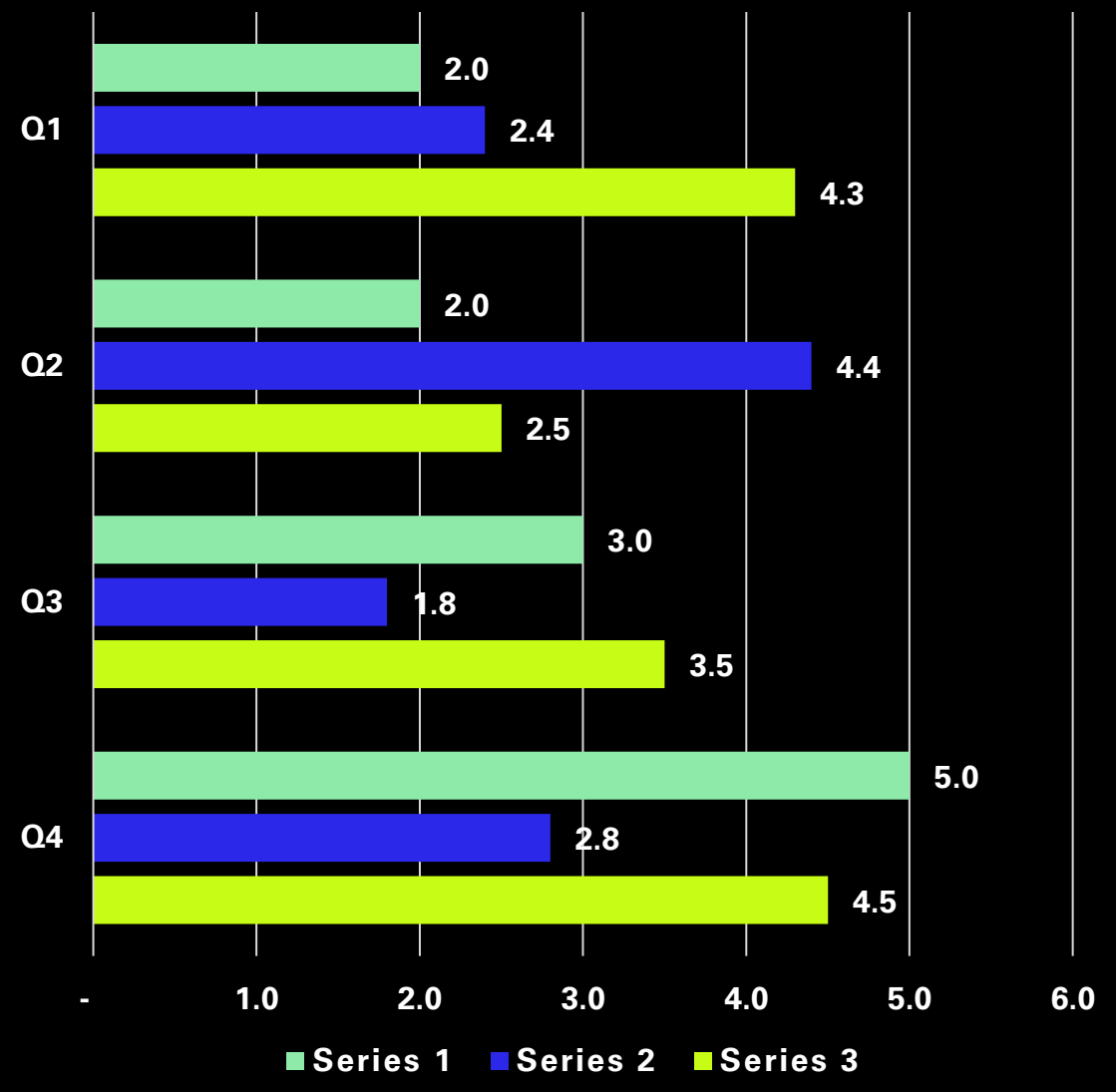
# PROBLEM STATEMENT

**ATHLETES AND TRAINERS FACE DIFFICULTY IN DETERMINING:**

WHETHER THE BODY HAS FULLY RECOVERED

WHEN TO CONTINUE TRAINING OR REST

HOW TO AVOID OVERTRAINING



# DATASET

The dataset contains athlete-related features:

**Input Features:**

- **Training Load** → intensity of workouts
- **Sleep Hours** → recovery time
- **Heart Rate** → physical stress indicator
- **Muscle Soreness** → fatigue level
- **Stress Level** → mental + physical stress

**Output:**

- **Recovery Status**
  - 1 → Recovered
  - 0 → Overtraining Risk

# Data Preprocessing

Before applying Machine Learning, data must be prepared.

Steps:

Data Cleaning

Remove missing or incorrect values

Normalization

Scale values for better performance

Feature Selection

Choose important features only

Train-Test Split

80% → Training data

20% → Testing data

# Logistic Regression

Used for **classification problems**

Predicts probability between **0 and 1**

👉 Example:

Output = 0.8 → Recovered

Output = 0.2 → Overtraining

**How it works:**

Draws a boundary between two classes

Uses a function called **sigmoid function**

**Advantages:**

Simple and fast

Works well for small datasets

# Decision Tree

Works like a **flowchart or tree structure**

👉 Example:

Is Sleep < 6 hours?

YES → Overtraining

NO → Check Stress Level

**How it works:**

Splits data based on conditions

Creates branches until decision is reached

**Advantages:**

Easy to understand

Good for decision-making

# Random Forest

Collection of **multiple decision trees**

👉 Instead of one tree → uses many trees

**How it works:**

Each tree gives a prediction

Final output = **majority voting**

**Advantages:**

More accurate

Reduces overfitting

Works well for complex data

# OUTPUT SCREEN SHOTS

ML_project.ipynb - Colab

colab.research.google.com/drive/1B2BEMQcCqV6wN3wjB9ntrdnDSw8vAv4J

```
# ----------------------------
# 📊 GRAPH 3: Confusion Matrix
# ----------------------------
cm = confusion_matrix(y_test, rf_pred)

plt.figure()
plt.imshow(cm)
plt.title("Confusion Matrix (Random Forest)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.savefig("confusion_matrix.png")
plt.show()
```

```
   Training_Load  Sleep_Hours  Heart_Rate  Muscle_Soreness  Stress_Level  \
0           39.9          6.0       152.0                3             4
1           46.7          8.3       156.0                3             1
2           94.1          7.6       166.0                3             2
3           35.2          7.0       169.0                3             3
4           72.1          5.2       169.0                3             4

  Recovery_Status
0       Recovered
1       Recovered
2       Recovered
3       Recovered
4       Recovered

Accuracy Results:
Logistic Regression: 0.9944444444444445
Decision Tree: 1.0
Random Forest: 1.0
```

# Model Training

Steps involved:
Load dataset
Split into training and testing
Train models (LR, DT, RF)
Predict results
The model learns patterns like:
High training + low sleep → Overtraining
Balanced training + good sleep → Recovered

# SUMMARY
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This project focuses on predicting **muscle recovery and overtraining risk** using Machine Learning. Athletes often face difficulty in knowing whether their body has fully recovered after workouts, which can lead to fatigue, poor performance, and injuries.

To solve this problem, a dataset containing features like **training load, sleep hours, heart rate, muscle soreness, and stress level** is used. The data is preprocessed and divided into training and testing sets.

# THANK YOU

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**B09**

https://github.com/haritejaathirajula-afk/Muscle_Recovery_-_Overtraining_Prediction